HISTONE FUNCTION

The histone H3-H4 tetramer is a copper reductase enzyme

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Eukaryotic histone H3-H4 tetramers contain a putative copper (Cu^{2+}) binding site at the H3-H3′ dimerization interface with unknown function. The coincident emergence of eukaryotes with global oxygenation, which challenged cellular copper utilization, raised the possibility that histones may function in cellular copper homeostasis. We report that the recombinant *Xenopus laevis* H3-H4 tetramer is an oxidoreductase enzyme that binds Cu^{2+} and catalyzes its reduction to Cu^{1+} in vitro. Loss- and gain-of-function mutations of the putative active site residues correspondingly altered copper binding and the enzymatic activity, as well as intracellular Cu^{1+} abundance and copper-dependent mitochondrial respiration and Sod1 function in the yeast *Saccharomyces cerevisiae*. The histone H3-H4 tetramer, therefore, has a role other than chromatin compaction or epigenetic regulation and generates biousable Cu^{1+} ions in eukaryotes.

Eukaryotic histones evolved from histone-like proteins of Archaea, single-celled organisms that lack nuclei and display genomes that are smaller than those of eukaryotes. Archaeal histones form a structure similar to the eukaryotic H3-H4 tetramer (*1*) but, unlike eukaryotic histones, typically lack extended N-terminal tails and posttranslational modifications. Because Archaea have no apparent capability for histone-based epigenetic regulation or requirement for genome compaction to fit within a nucleus, this prompted us to consider whether the conserved histone H3-H4 tetramer may have an additional function.

The dimerization interface of the two histone H3 proteins contains cysteine and histidine residues (Fig. 1A) in an arrangement typical of Cu^{2+} binding sites in other proteins (*2*). The evolutionary conservation of residues in this region is greater than expected from their contributions to nucleosome stability (*3*), consistent with a potential metal binding capability (*4*–*6*).

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Eukaryotes emerged about the same time as the initial accumulation of oxygen (*7*). This oxygenation event decreased the biological usability and increased the toxicity of transition metals such as copper (*8*). We hypothesize that histones may have facilitated metal utilization in oxidizing conditions.

Copper often serves as a redox cofactor for enzymes such as cytochrome c oxidase in the mitochondrial electron transport chain (ETC) and Cu, Zn-superoxide dismutases (e.g., Sod1) (*9*). As a redox cofactor, copper cycles between the cupric (Cu^{2+}) and cuprous (Cu^{1+}) states. However, it is the cuprous form that is trafficked and sensed intracellularly, suggesting a need to maintain cellular copper in the Cu^{1+} oxidation state for proper distribution and utilization (*10*). Whether protein-based mechanisms have evolved to regulate intracellular Cu^{2+} reduction is unknown. Here, we present evidence that the eukaryotic H3-H4 tetramer binds Cu^{2+}, catalyzes Cu^{2+} reduction, and improves the utilization of copper by nuclear, cytoplasmic, and mitochondrial proteins.

Recombinant histone H3-H4 tetramer binds cupric ions

To determine whether copper ions interact with the residues at the H3-H3′ interface (*4*) (Fig. 1, A and B), we assembled and purified recombinant *Xenopus laevis* wild-type (Xl WT) histone H3-H4 tetramers (*11*) (fig. S1, A to C). Incubation of the Xl WT tetramer with increasing amounts of Cu^{2+} in a Tricine buffer resulted in an ultraviolet (UV) absorbance band that is characteristic of copper-cysteine interactions (*12*) (Fig. 1C and fig. S1D). UV absorbance plateaued at a molar ratio of ~1, suggesting a stoichiometry of one copper ion per tetramer (Fig. 1C and fig. S1D). Mutation of Cys^{110} (C110) to alanine (*H3C110A*) (fig. S1, B and C) abolished this copper-dependent absorbance change (Fig. 1C). Isothermal titration calorimetry (ITC) analysis further demonstrated high affinity of the tetramer for Cu^{2+} and an approximately equimolar stoichiometry (Fig. 1D). The WT H3-H4 complex also exhibited greater retention on a Cu^{2+} affinity column than the $H3^{C110A}$ counterpart (fig. S1E).

The histone H3-H4 tetramer catalyzes cupric ion reduction

We next hypothesized that a mechanism to maintain the Cu^{1+} state intracellularly may be beneficial in oxidizing conditions. Thus, we asked whether the H3-H4 tetramer catalyzes the reduction of Cu^{2+} to Cu^{1+} in the presence of a source of electrons. We developed an assay using the chelators neocuproine (NC) or bicinchoninic acid (BCA) to detect Cu^{1+} production spectrophotometrically (Fig. 2A and fig. S2A). Assays contained reduced forms of either tris(2-carboxyethyl)phosphine (TCEP), nicotinamide adenine dinucleotide (NADH), or its phosphate form (NADPH) as electron donors. Reactions were initiated by addition of Cu^{2+} in the form of either $CuCl_2$-Tricine, $CuCl_2$-N-(2-acetamido)iminodiacetic acid (ADA), or $CuCl_2$-nitrilotriacetic acid (NTA) solutions (fig. S2B). Spontaneous Cu^{1+} production occurred at a slow rate, but the rate of Cu^{1+} production substantially increased in the presence of the Xl WT tetramer (Fig. 2B and fig. S2, C to E). Production of Cu^{1+} eventually plateaued owing to near full consumption of the electron source (Fig. 2B). No appreciable production of Cu^{1+} occurred in the absence of TCEP, indicating that it, and not the tetramer, was the source of electrons. The tetramer enhanced Cu^{2+} reduction in anaerobic conditions as well, ruling out any role for oxygen (fig. S2F). The Xl WT tetramer assembled into a complex with the C terminus of the human histone chaperone Spt2 (*13*) also enhanced Cu^{2+} reduction (fig. S2, G to I). Additionally, the tetramer could use natural reductants such as NADPH to increase Cu^{2+} reduction (fig. S3A). Direct measurement of NADPH confirmed its oxidation during Cu^{2+} reduction (fig. S3B). The effect of the tetramer was specific to cupric ions because there was no enhancement of ferric ion reduction.

Heating (Fig. 2B) or proteinase K digestion (fig. S3C) of the Xl WT tetramer abolished Cu^{2+} reduction activity. The unassembled Xl histone H3 or the yeast H2A also did not enhance Cu^{2+} reduction (fig. S3, C and D). Increasing rates of Cu^{1+} production occurred with increasing amounts of either TCEP (Fig. 2C) or Cu^{2+} (Fig. 2D), eventually approaching a maximum rate, consistent with protein-based catalysis (fig. S3, E and F).

The *H3C110A* mutation substantially reduced the rate of Cu^{2+} reduction (Fig. 2E), as did treatment with *N*-ethylmaleimide (NEM) (fig. S3G). Ribonuclease A (RNase A), which

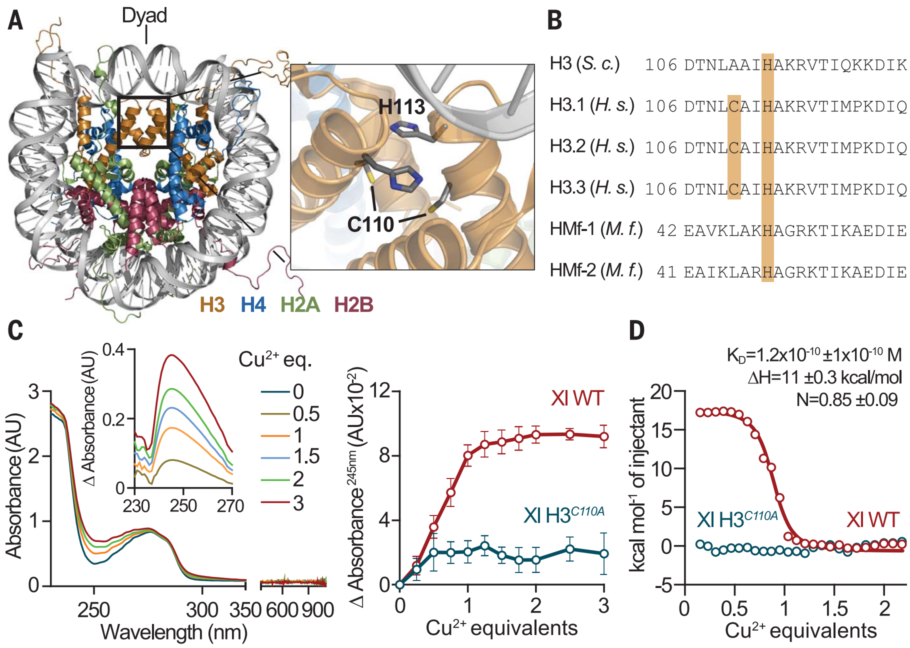

Fig. 1. Recombinant *X. laevis* histone H3-H4 tetramer interacts with cupric ions. (**A**) Left: *X. laevis* (Xl) nucleosome core particle structure [Protein Data Bank (PDB) 1KX5] (*38*). The box delineates the H3-H3′ interface. Right: Interface residues H3H113 and H3C110 are shown. (**B**) Alignment of the C-terminal region of *S. cerevisiae* (*S. c.*) and *Homo sapiens* (*H. s.*) histone H3 and archaeal [*Methanothermus fervidus* (*M. f.*)] histones. (**C**) Left: UV-visible absorbance spectrum of the Xl H3-H4 tetramer incubated with or without Cu^{2+}. Inset: Differential absorbance compared to tetramer without Cu^{2+}. Right: Buffer-corrected differential absorbance of the indicated Xl tetramers. AU, absorbance units; eq., equivalents. (**D**) Representative ITC titration profile of the Xl H3-H4 tetramer. Circles are experimental data, and the line is the fitted curve. Average dissociation constant (K_D), enthalpy change (ΔH), and stoichiometry (N) ± SD of the H3-H4 tetramer-Cu^{2+} complex calculated from three experiments are shown. Single-letter abbreviations for the amino acid residues are as follows: A, Ala; C, Cys; D, Asp; E, Glu; F, Phe; G, Gly; H, His; I, Ile; K, Lys; L, Leu; M, Met; N, Asn; P, Pro; Q, Gln; R, Arg; S, Ser; T, Thr; V, Val; W, Trp; and Y, Tyr.

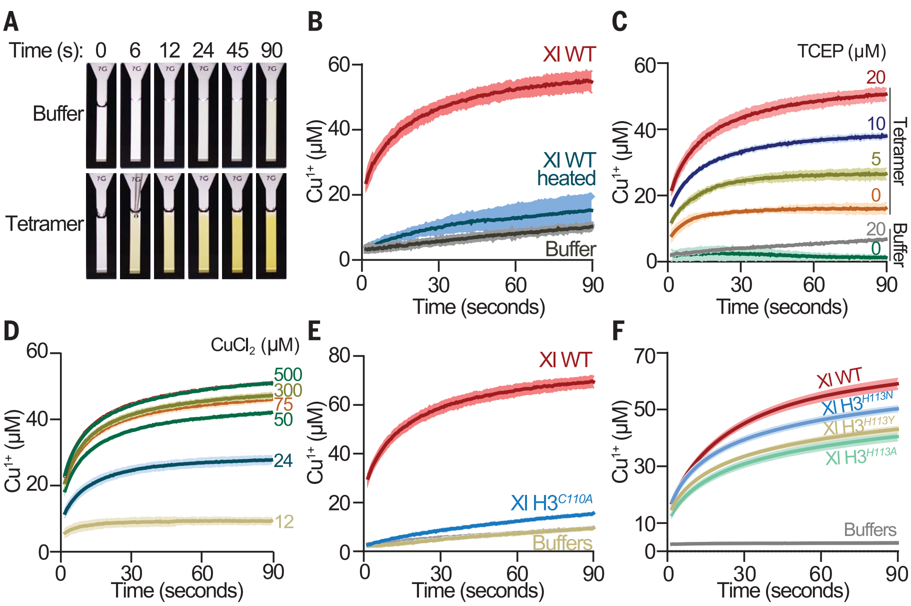

Fig. 2. The *X. laevis* H3-H4 tetramer catalyzes reduction of cupric ions. (**A**) Photographic representation of in vitro Cu^{2+} reduction assay. The yellow color is due to NC_2-Cu^{1+} complex formation. (**B**) Progress curves of Cu^{2+} reduction with 1 μM of tetramer, heated tetramer, or buffer in presence of 30 μM TCEP and 1 mM $CuCl_2$-10 mM Tricine. Lines and shading represent the mean ± SD of three to five assays. (**C** to **F**) Same as (B) but with the indicated amount of (C) TCEP or (D) $CuCl_2$; or with (E) 1 μM of the indicated tetramers or (F) 5 μM of the indicated tetramers in the presence of 100 μM TCEP and 0.5 mM $CuCl_2$-ADA pH 8.

contains redox-reactive cysteines (*14*), did not enhance Cu^{2+} reduction (fig. S3H). Similarly, dithiothreitol (DTT) directly reduced an equimolar amount of Cu^{2+} but did not substantially enhance the rate of Cu^{2+} reduction by TCEP (fig. S3H). These data underscore the importance of H3C110 to the tetramer enzyme activity.

Mutation of H3 His[113] (H3H113) (*4*, *5*) to alanine (*H3H113A*), or to asparagine or tyrosine (*H3H113N* or *H3H113Y*), which have been found in certain cancers (*15*), had little effect on the rate of reduction with Cu^{2+}-Tricine as the substrate but diminished Cu^{2+} reduction with Cu^{2+}-ADA (Fig. 2F and fig. S2B) or Cu^{2+}-NTA as substrates (figs. S2B and S3I), likely because of the differing Cu^{2+} coordination ability of different buffers. Altogether, our data reveal that the H3-H4 tetramer catalyzes Cu^{2+} reduction by various electron donors and is thus a cupric reductase.

The yeast histone H3 (yH3) contains H113 but lacks the C110 residue found in most other eukaryotic H3s (Fig. 1B). Consistently, recombinant yH3-H4 tetramer did not absorb UV light in the 245-nm range when incubated with Cu^{2+}. However, a broad peak centered at 680 nm was observed at high concentrations of yeast tetramer and high ionic strength (2 M NaCl), consistent with weakly absorbing d-d transitions typical of coordinated Cu^{2+} ions (*16*) (Fig. 3A and fig. S4A). The *H3H113A* mutation (fig. S4, B and C) decreased the copper-dependent absorbance change at 680 nm (Fig. 3A). Furthermore, retention of the yH3-H4 complex on a Cu^{2+} affinity column was reduced by the *H3H113A*, *N*, or *Y* mutations (fig. S4, B to F), consistent with a role of H3H113 in Cu^{2+} interaction.

Mutation of H3A110 to cysteine (yH3^{A110C} tetramer) (fig. S4, G and H) elicited a copper-dependent UV absorbance band nearly identical to that observed with the Xl WT tetramer (Fig. 3B). Furthermore, the *H3H113A* mutation of the yH3^{A110C} tetramer (i.e., yH3^{A110CH113A} tetramer) (fig. S4, G to I) diminished copper-dependent absorbance at 245 nm (Fig. 3B), indicating that H3H113 affects the cysteine-Cu^{2+} interaction. The yH3^{A110C} tetramer also displayed cupric reductase activity, which was diminished by the *H3H113A* mutation (Fig. 3C). In the absence of H3C110, the yeast tetramer did not display cupric reductase activity in standard salinity, likely because of deficient Cu^{2+} binding, which required 2 M NaCl (Fig. 3A). High salinity interfered with other assay components, precluding further investigation. Nonetheless, our findings suggest that the yH3-H4 tetramer can display cupric reductase activity and that H3H113 participates in this function.

Mutation of histone H3H113 results in loss of function in yeast

We next investigated whether mutation of the H3H113 residue in vivo would diminish the Cu^{1+} pool. Mutation of H3H113 to alanine is lethal in *Saccharomyces cerevisiae* (*17*) (fig. S5A). However, introduction of either *H3H113N* or *H3H113Y* mutations in the two chromosomal copies of the histone H3 gene generated viable and somewhat slow-growing yeast strains *H3*^{H113N} and *H3*^{H113Y} (fig. S5B). Yeast strains in which the *H3H113N* and *H3H113Y* mutant histones were expressed from one locus, with the other locus deleted, were severely sick compared with strains expressing both histone loci (fig. S5C). This finding identifies *H3H113N* and *H3H113Y* as loss-of-function alleles, coincident with the loss of Cu^{2+} binding in vitro. We mostly used the *H3H113N* mutation in genetic experiments because *H3H113Y* results in a more severe growth defect, potentially confounding the interpretation of phenotypes.

H3H113 mutations alter cuprous ion availability

We next assayed the activity of the transcription factor Mac1 as a readout of nuclear Cu^{1+} abundance because it is directly inhibited by Cu^{1+} and does not bind Cu^{2+} (*18*). Gene expression analysis revealed largely similar profiles in WT and *H3*^{H113N} strains in rich fermentative media [synthetic complete (SC) and yeast extract peptone dextrose (YPD)] (fig. S6A). However, Mac1 target genes were up-regulated in both the *H3*^{H113N} (fig. S6B) and *H3*^{H113Y} strains (Fig. 4A). Deletion of *CTR1* (*ctr1*Δ), the main copper importer in yeast (*19*), increased Mac1 target expression (Fig. 4B and fig. S6, B and C). However, Mac1 targets displayed even greater up-regulation and incomplete repression in response to exogenous copper in *H3*^{H113N}*ctr1*Δ (Fig. 4B and fig. S6C). *ctr1*Δ substantially reduced total copper concentrations, whereas the *H3H113N* mutation did not (Fig. 4C), suggesting that the H3H113 mutations decrease Cu^{1+} abundance without affecting total copper concentration.

We developed a reporter plasmid to assess Cu^{1+} abundance by employing the Cup2 transcription factor, which is directly activated by Cu^{1+} but not Cu^{2+} (*20*). The plasmid harbors the green fluorescent protein (*GFP*) gene downstream of the *CUP1* promoter, which is the main target of Cup2 (*21*) (Fig. 4D and fig. S6D). GFP expression was proportional to the expected Cu^{1+} abundance in both physiological and genetic assays (Fig. 4E and fig. S6, E and F) and was reduced in *H3*^{H113N} (Fig. 4F and fig. S6G), further indicating that H3H113 is required to maintain cellular Cu^{1+} availability.

The H3H113 residue supports copper utilization for mitochondrial respiration

We next asked whether the H3H113 mutations impair copper utilization outside the nucleus. We first tested mitochondrial respiration by assessing copper-dependent cytochrome c oxidase function. Although *H3*^{H113N} did not display a defect in mitochondrial respiration, the *H3*^{H113Y} strain displayed a significant loss of O_2 consumption (Fig. 5A and fig. S7A). This defect was exacerbated in the presence of the copper chelator bathocuproinedisulfonic acid (BCS) and was recovered by excess exogenous copper (fig. S7, B and C).

To relate the defects in mitochondrial respiration to disruption of copper utilization, we assessed the ability of cells to increase respiratory function as they were shifted from a copper-depleted state to a copper-replete state. Copper depletion by means of *CTR1* deletion resulted in a low rate of antimycin-sensitive O_2 consumption (Fig. 5B and fig. S7D) and cytochrome c oxidase activity (Fig. 5C), which were

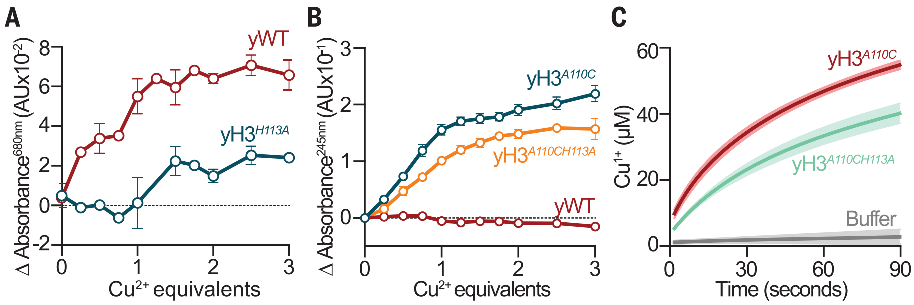

Fig. 3. The yeast H3-H4 tetramer potentially is a copper reductase. (**A** and **B**) Buffer-corrected differential absorbance of the indicated yeast tetramers. (**C**) Progress curves of Cu^{2+} reduction with 5 μM of tetramers in the presence of 100 μM TCEP and 0.5 mM $CuCl_2$-ADA pH 8.

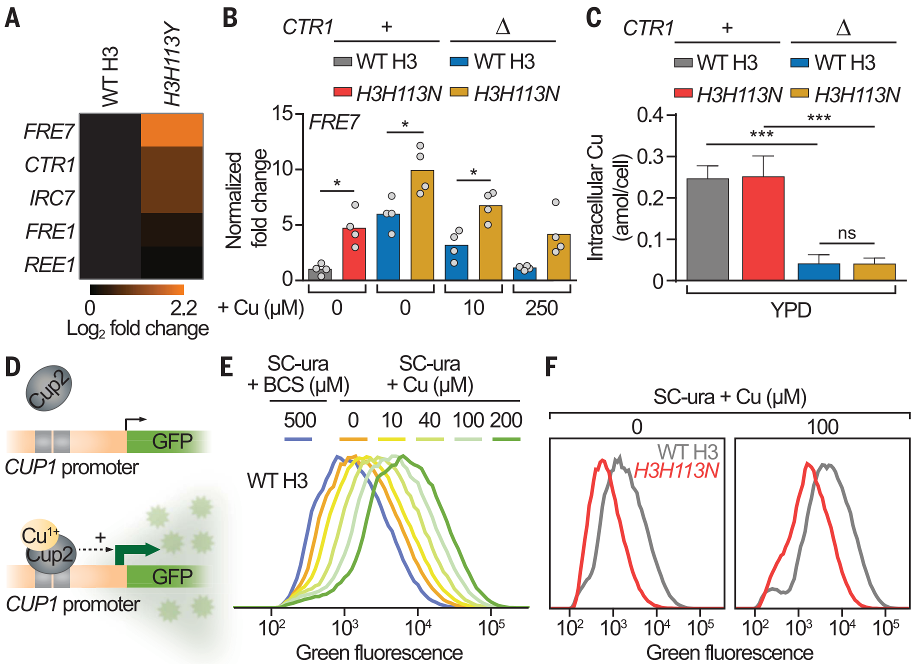

Fig. 4. H3H113 regulates Cu^{1+}-dependent transcriptional activities of Mac1 and Cup2. (**A** and **B**) Reverse transcription quantitative polymerase chain reaction (RT-qPCR) analyses of Mac1 target gene (*39*) expression relative to WT and normalized to *ACT1* expression. The (A) boxes and (B) bars show means from four independent experiments (dots) in YPD with or without $CuSO_4$. Baseline copper concentration in YPD is ~1 μM. (**C**) Intracellular copper content of exponentially growing strains. Bars are means ± SD from three to six experiments. (**D**) Schematic representation of the p$^{(CUP1)}$-GFP reporter system. (**E** and **F**) Average flow cytometry distributions of cells containing the p$^{(CUP1)}$-GFP plasmid grown in SC lacking uracil (SC-ura) with or without BCS or $CuSO_4$ from five or six experiments. Baseline copper concentration in SC is ~0.25 μM. *$P \le 0.05$; ***$P \le 0.001$; ns, not significant.

gradually increased by addition of exogenous copper. However, substantially more copper was required to rescue *ctr1*Δ in the context of *H3H113N* (Fig. 5, B and C). More copper was also required to rescue respiratory growth of *ctr1*Δ in the context of *H3H113N* or *H3H113Y* (Fig. 5D and fig. S7E). *H3H113N* also diminished the copper-dependent rescue of cells lacking *MAC1*, which activates *CTR1* expression (*18*) (fig. S7F). Addition of iron, zinc, or manganese did not rescue the respiratory growth defects of *ctr1*Δ strains (fig. S7G).

Consistent with the hypomorphic nature of *H3H113N*, a strain with one WT H3 and one *H3H113N* gene displayed an intermediate defect compared with strains containing two *H3H113N* or two WT H3 genes (fig. S8A). Similarly, deleting one of the two copies of the H3 and H4 genes (*hht1-hhf1*Δ) (fig. S8, B to D) increased the copper requirement in *ctr1*Δ for respiratory growth (fig. S8E).

Total copper and iron concentrations were similar between *H3^H113N^* and WT in respiratory medium (Fig. 5E and fig. S9A). Addition of excess copper increased total copper concentrations similarly in both strains (Fig. 5E), indicating that the copper utilization defect in *H3^H113N^ctr1*Δ was not due to changes in total copper abundance. Inefficient recovery of *ctr1*Δ in the context of *H3H113N* was not due to increased Cu^{1+} sequestration by the metallothionein Cup1, because loss of Cup1 (*cup1^F8stop^*) had no effect on the requirement for copper in *H3^H113N^ctr1*Δ (fig. S9B).

We considered whether potential disruptions in chromatin accessibility or gene regulation account for the copper utilization defects of the H3H113 mutants. The *H3H113N* mutation or deletion of Asf1, a histone chaperone that facilitates nucleosome assembly in part through interaction with H3H113 (*22*, *23*), resulted in minimal disruption of chromatin accessibility (fig. S9C). However, *asf1*Δ did not phenocopy the H3H113 mutants in respiratory media (fig. S9D). Global gene expression patterns were similar between WT and *H3^H113N^* strains in respiratory media (fig. S9E), with comparable up-regulation of genes involved in respiratory growth and copper regulation (fig. S9, F and G).

Deletion of membrane-bound metalloreductases *FRE1* and *FRE2* (*24*) did not phenocopy the H3H113 mutations (fig. S10A), suggesting that the intracellular role of histone H3 on copper utilization is distinct from extracellular metal reduction. Glutathione (GSH) also participates in cellular copper metabolism (*25*). Decreasing GSH abundance, by deleting the *GSH1* gene, increased the amount of copper required to rescue *ctr1*Δ but to a lesser extent than *H3H113N* (fig. S10B). Combining *H3H113N* with *gsh1*Δ caused an even greater defect in copper utilization (fig. S10B), suggesting that histones have a distinct role in copper utilization.

Histone H3 supports copper utilization for Sod1 function

We next considered whether histone H3 might also affect copper utilization by Sod1. Total Sod1 activity was ~40% less in *H3^H113Y^* compared with WT, which was recovered by excess exogenous copper (fig. S11, A and B). *H3H113N* also decreased Sod1 activity by ~20% (Fig. 5F). Deletion of *CTR1* reduced Sod1 activity and formation of its internal disulfide bond, which are dependent on Cu^{1+} and the copper chaperone Ccs1 (*26*) (Fig. 5F). Combining *ctr1*Δ with *H3H113N* further decreased the internal disulfide bond and Sod1 activity, which were restored by addition of excess exogenous copper (Fig. 5F). Loss of function of Sod1 causes lysine auxotrophy in yeast (*27*). Correspondingly, *H3^H113N^ctr1*Δ exhibited a growth defect in lysine-deficient conditions (fig. S11C).

Deletion of *CCS1* substantially decreased Sod1 activity (Fig. 5G), which was restored by addition of excess copper, but the *H3H113N* mutation substantially increased the amount

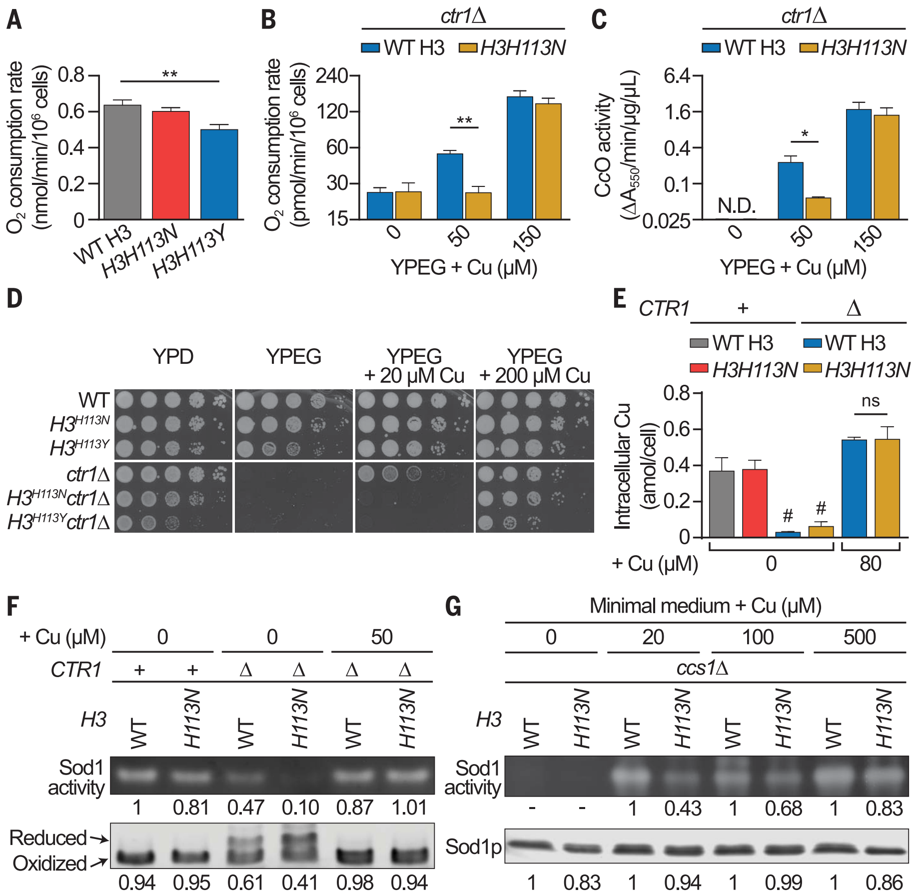

Fig. 5. H3H113 is required for utilization of copper for mitochondrial respiration and Sod1 function. (**A** and **B**) Oxygen consumption assays of cells incubated for (A) 18 or (B) 4 hours in liquid YPEG (yeast extract peptone ethanol glycerol) with or without $CuSO_4$. Baseline copper concentration in YPEG is ~1 μM. Bars show means in (A) linear and (B) $\log_2$ scale ± SD from three experiments. Bars in (A) are scaled to mitochondrial DNA contents. (**C**) Cytochrome c oxidase assays of cells incubated for 4 hours in liquid YPEG with or without $CuSO_4$. Bars show means in $\log_2$ scale ± SD from three experiments. N.D., not detectable. (**D**) Spot test assays in media with or without $CuSO_4$. (**E**) Intracellular copper content of cells grown in YPEG with or without $CuSO_4$. Bars show means ± SD from three to six experiments. #The *ctr1*Δ strains, which do not grow in nonfermentable media, were incubated in YPEG for 12 hours and assessed for metal content for reference. (**F**) Representative Sod1 activity (top) and Sod1 disulfide-bond assays (bottom) from three experiments for cells grown in SC with or without $CuSO_4$. Relative signal intensities are indicated (bottom numbers are the ratio of oxidized to total Sod1). (**G**) Same as (F) but for cells grown in minimal medium with or without $CuSO_4$. Baseline copper concentration in minimal medium is ~0.25 μM. *$P \le 0.05$; **$P \le 0.01$; ns, not significant.

of copper required for recovery (Fig. 5G). Considerably more copper was also required to rescue the lysine auxotrophy of *ccs1Δ* in the context of *H3H113N* (fig. S11, D and E). Unlike the effect of *H3H113N*, deletion of *CTR1* did not increase the requirement of *ccs1Δ* for exogenous copper (fig. S11F), suggesting that histones contribute to copper utilization differently than Ctr1. Excess copper did not rescue the lysine auxotrophy of *sod1Δ* strains (fig. S11G), whereas hypoxia rescued the lysine auxotrophy of *ccs1Δ* and *sod1Δ* regardless of H3H113 (fig. S11H), indicating that H3H113 is relevant to this phenotype when Sod1 function is required.

Addition of manganese, zinc, or iron did not rescue *ccs1Δ* lysine auxotrophy (fig. S11I). The increased copper requirement of *H3^{H113N}ccs1Δ* was not due to differences in Cup1-dependent Cu^{1+} sequestration (fig. S12A), intracellular copper and iron concentrations (fig. S12, B and C), or Asf1 (fig. S11E). Gene expression differences also did not account for defective copper utilization because *ccs1Δ* and *H3^{H113N}ccs1Δ* displayed similar patterns (fig. S12, D to F). Altogether, our findings reveal that the H3H113 residue is also important for copper utilization by Sod1. The considerable impacts of H3H113 mutations on Cu^{1+} abundance and on at least three separate copper-dependent functions support the model that the cupric reductase function of the H3-H4 tetramer regulates Cu^{1+} availability in yeast.

The *H3A110C* mutation enhances cuprous ion availability and copper utilization

The in vitro copper reductase results suggested that *H3A110C* might serve as a gain-of-function mutation to increase Cu^{1+} and improve copper utilization in vivo. The yeast *H3^{A110C}* strain grew similarly to WT cells (fig. S13A) but enhanced the ability of exogenous copper to restore the respiratory growth of *ctr1Δ* (Fig. 6A). We reasoned that increased provision of Cu^{1+} might mitigate the depletion of cellular GSH. Indeed, *H3^{A110C}gsh1Δ* grew better than *gsh1Δ* in copper-depleted respiratory conditions (fig. S13B) with a greater O_2 consumption rate (Fig. 6B and fig. S13C). *H3^{A110C}* exhibited a copper-dependent growth advantage in the presence of a sublethal amount of potassium cyanide, an inhibitor of cytochrome c oxidase, independently of GSH abundance (Fig. 6C and fig. S13D). Correspondingly, the *H3^{A110C}gsh1Δ* strain displayed greater Cup2 activity than *gsh1Δ* (Fig. 6D), indicating increased Cu^{1+} abundance. Additionally, the *H3A110C* mutation enhanced the copper-dependent rescue of lysine auxotrophy in *ccs1Δ* (Fig. 6E). The *H3A110C* mutation even rescued the diminished Cup2 activity of *H3^{H113N}* (fig. S13E), the copper utilization defect of the *H3^{H113N}ctr1Δ* strain (Fig. 6F), and the lethality of the *H3H113A* mutation (Fig. 6G), corroborating the opposing effects of the H3A110 and H3H113 mutations both in vivo and in vitro. These results highlight the advantage in copper utilization conferred by the *H3A110C* mutation and further support that the yeast histone H3 regulates the Cu^{1+} pool and copper utilization.

Discussion

Histones have been known as packaging and regulatory proteins for the eukaryotic genome. We now reveal that the histone H3-H4 tetramer is also a cupric reductase that provides usable Cu^{1+} for the cell. The enzymatic activity suggests that the protein complex has many uncharacterized features, including a catalytic site at the H3-H3′ interface, that contribute to Cu^{2+} binding, increasing its reduction potential and promoting electron transfer (*28*).

The H3-H3′ interface forms in vivo mostly during nucleosome assembly. Therefore, the commencement of enzymatic activity may be coupled to the protection of DNA as it wraps around the nucleosome. Such a coupling may decrease the toxicity of Cu^{1+} by the very enzymes that produce it as a beneficial adaptation in species that require its intracellular use. However, proteins such as Spt2 (*13*) may stabilize the H3-H3′ interface outside of the nucleosomal

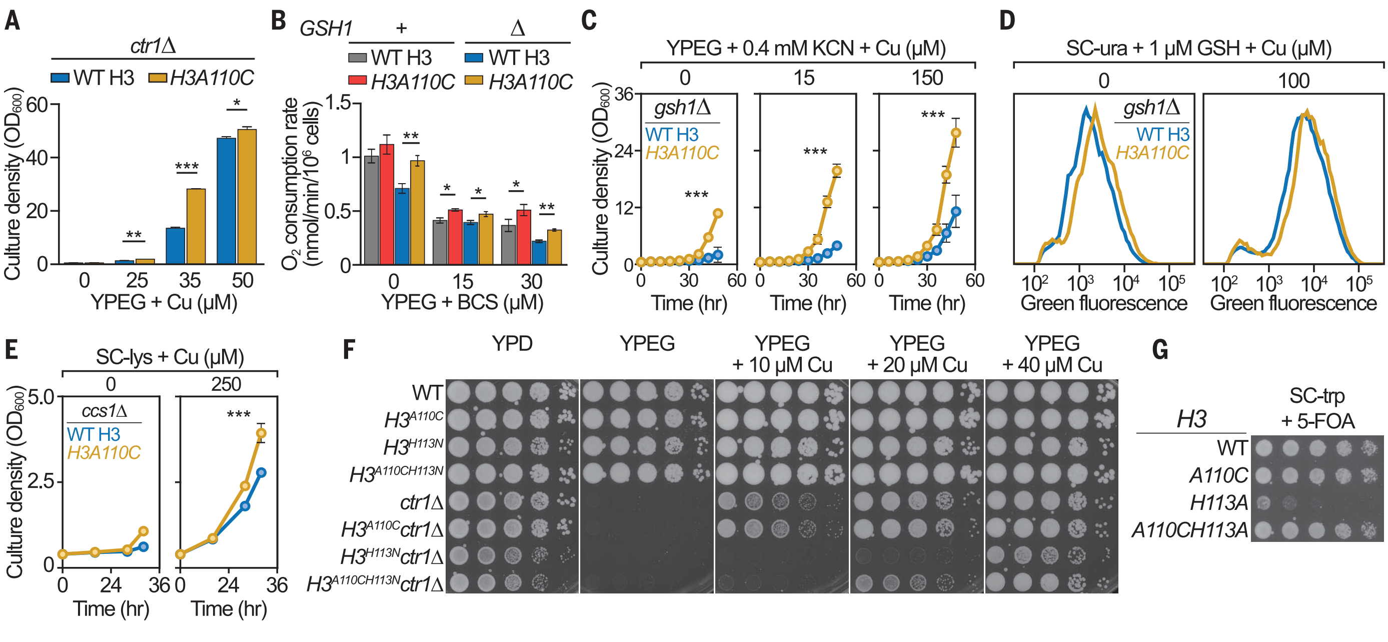

Fig. 6. The *H3A110C* mutation enhances copper utilization in *S. cerevisiae*. (**A**) Growth after 48 hours in liquid YPEG with or without $CuSO_4$. Bars show mean optical density at a wavelength of 600 nm (OD_{600}) ± SD from three experiments. (**B**) Oxygen consumption assays of cells grown in the indicated media for 12 hours. Bars show mean ± SD of three experiments and are scaled to mitochondrial DNA contents. (**C**) Growth curves in YPEG with potassium cyanide (KCN) with or without $CuSO_4$. Lines show means at each time point ± SD from three experiments. *P* value is based on all time points. (**D**) Average flow cytometry distributions of cells containing the $p^{(CUP1)}$-GFP plasmid grown in liquid media with or without $CuSO_4$ from three experiments. (**E**) Same as (C) but for cells grown in SC lacking lysine (SC-lys) with or without $CuSO_4$ from four experiments. (**F**) Spot test assays in media with or without $CuSO_4$. (**G**) Plasmid shuffle assay with strains harboring WT H3 on a URA3 plasmid and the indicated H3 gene on a TRP1 plasmid. 5-fluoroorotic acid (5-FOA) is lethal to cells that cannot lose the URA3 plasmid. $*P < 0.05$; $**P \le 0.01$; $***P \le 0.001$.

context, affecting regulation and timing of enzymatic activity relative to other chromatin-based events.

The mechanism of electron transfer by the tetramer is unclear but may involve direct transfer from reductants to Cu^{2+} at the active site or through a chain of residues from a distant site (*29*, *30*). Copper is present in the nucleus (*31*, *32*) but is likely bound (*33*) and transported to and from histones through as-yet-to-be-identified chaperones, possibly including Atox1 (*34*). Copper disproportionately accumulates in the nucleus when its cellular efflux is compromised, such as in Wilson's disease (*35*, *36*), suggesting that the nucleus is a major transit hub for cellular copper.

The oxidoreductase function of the H3-H4 tetramer provides a reasonable hypothesis for why the archaeal ancestor of the eukaryotes possessed histone tetramers. Such enzymatic activity, with an associated intracellular Cu^{1+} transit system, could have helped maintain a functioning ETC in the proto-mitochondria, thereby making the presence of histones in the ancestral archaeon not incidental (*37*), but rather essential for eukaryogenesis.

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SUPPLEMENTARY MATERIALS

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Materials and Methods
Figs. S1 to S13
Table S1
References (*40–56*)
MDAR Reproducibility Checklist

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Science

The histone H3-H4 tetramer is a copper reductase enzyme

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Enzymatic activity of histones
Eukaryotic histones serve as structural elements to package DNA. However, they contain a copper-binding site for which the biological relevance is unknown. Copper homeostasis is critical for several fundamental eukaryotic processes, including mitochondrial respiration. Attar *et al.* hypothesized that histones may play a critical role in cellular copper utilization (see the Perspective by Rudolph and Luger). Using a multifaceted approach ranging from in vitro biochemistry to in vivo genetic and molecular analyses, they found that the histone H3-H4 tetramer is an oxidoreductase enzyme that catalyzes reduction of cupric ions, thereby providing biologically usable cuprous ions for various cellular processes. This work opens a new front for chromatin biology, with implications for eukaryotic evolution and human biology and disease.

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